# CHAPTER 3: RESULTS

## 3.1 Demographic and Occupational Profile

A total of 376 wood industry workers participated in the study, representing an 87.5% response rate from 430 initially approached. The mean age of participants was 34.69 ± 11.46 years (range: 18–60 years), and the mean duration of employment in the wood industry was 11.74 ± 9.67 years, indicating a sample that ranged from relatively recent entrants to seasoned professionals with substantial cumulative occupational exposure.

**Table 3.1: Socio-demographic and Occupational Characteristics of Study Participants**

| Variable | Category | n | Percentage % |
|---|---|---|---|
| Smoking status | Yes | 91 | 24.2 |
| | No | 285 | 75.8 |
| Education | Illiterate | 95 | 25.3 |
| | Primary | 113 | 30.1 |
| | Secondary | 146 | 38.8 |
| | Higher | 19 | 5.1 |
| Marital Status | Single | 104 | 27.7 |
| | Married | 272 | 72.3 |
| PPE usage | Never | 124 | 33.0 |
| | Sometimes | 161 | 42.8 |
| | Always | 91 | 24.2 |
| Workplace ventilation | Present | 315 | 83.8 |
| | Absent | 59 | 15.7 |

The majority of participants had attained modest levels of formal education: 38.8% had completed secondary schooling, 30.1% primary education, 25.3% were entirely illiterate, and a mere 5.1% had achieved higher education. This educational profile carries direct occupational safety implications, as limited literacy may impair workers' capacity to interpret hazard warning labels, follow written safety protocols, and engage effectively with formal training programs. Approximately one in four workers (24.2%) were current smokers — a proportion of particular significance given the synergistic respiratory risks conferred by simultaneous tobacco and wood dust exposure.

Regarding institutional safety infrastructure, 58.8% of workers reported having received formal occupational safety training, and 58.0% had been provided with PPE. However, actual PPE utilization was markedly inconsistent: only 24.2% reported always using protective equipment, 42.8% used it sometimes, and 33.0% never used it. This pronounced compliance gap between provision and consistent utilization represents a critical programmatic finding. It highlights that physical availability alone cannot overcome behavioral or ergonomic barriers to safety compliance, such as equipment discomfort or a workplace culture that normalizes risk-taking . The majority of workplaces (83.8%) were reported to have some form of ventilation system, while 15.7% of workers operated in poorly ventilated environments. For these individuals, the absence of primary engineering controls maximizes continuous breathing-zone exposure to suspended particulates, directly elevating their localized occupational risk.

the demographic profile of this cohort highlights a profound socioeconomic vulnerability that compounds their occupational health risks. The intersection of limited formal education, informal employment arrangements, and hazardous working conditions creates a multi-dimensional burden for these laborers. Notably, the financial precarity of this demographic is starkly reflected in their healthcare financing; a majority of the workforce (52.4%) bears the cost of medical treatment entirely out-of-pocket, with only a negligible fraction (8.2%) possessing formal health insurance. In an environment where over half the workforce has required time off due to occupational trauma, this lack of financial protection creates a vicious cycle where workplace morbidity directly precipitates economic hardship. Laborer continue to operate heavy machinery and endure particulate exposure despite actively suffering from chronic musculoskeletal pain or respiratory distress, simply out of financial necessity. Ultimately, this demographic snapshot reveals a workforce that is not only highly exposed to industrial hazards, but also systematically underequipped both educationally and economically to advocate for safer conditions or access restorative healthcare.

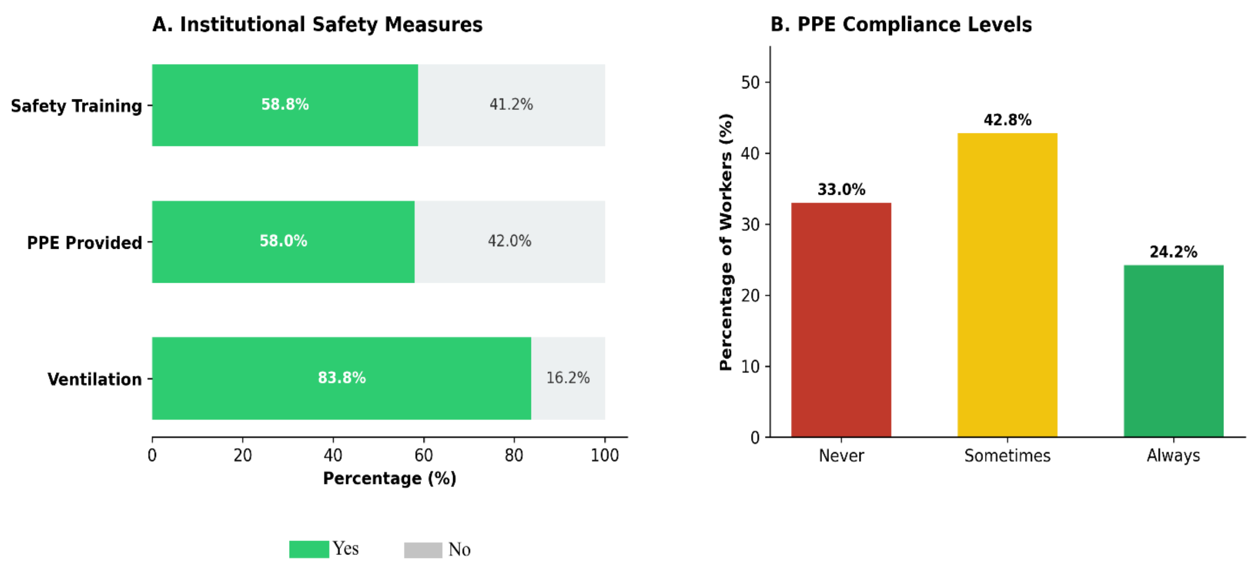


**Figure 2:** Institutional Safety and PPE Compliance

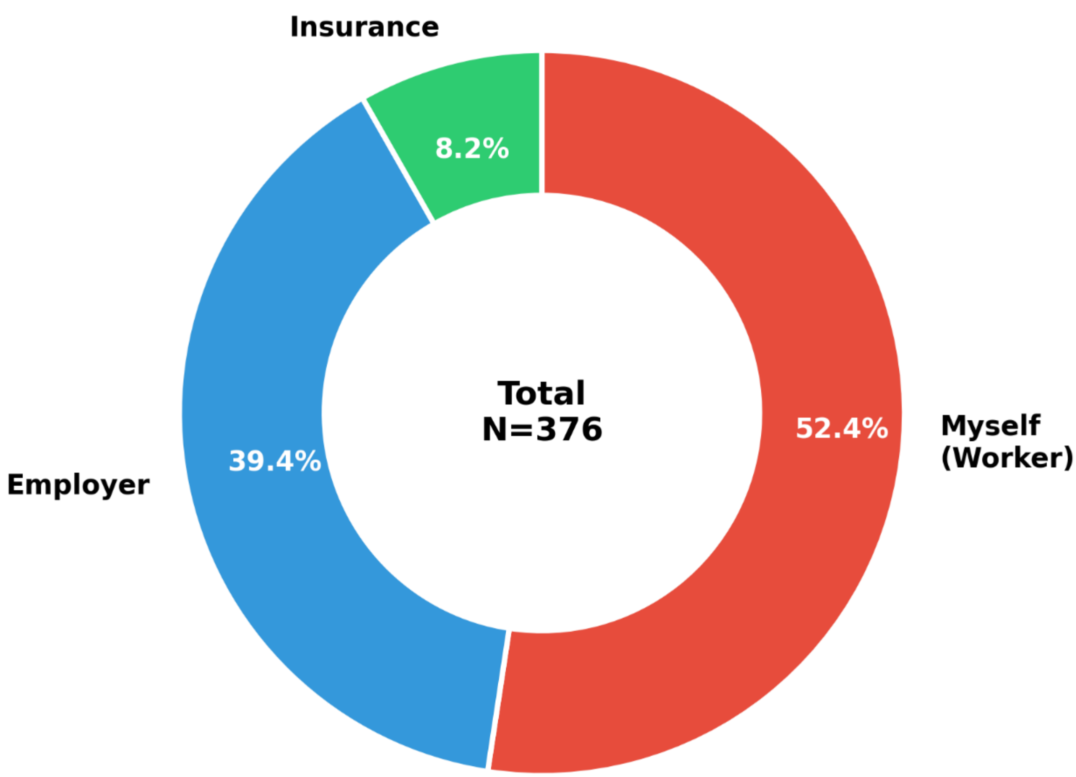


**Figure 3:** Medical Treatment Coverage

Notably, 52.4% of workers bear medical expenses personally, 39.4% are covered by employers, and only 8.2% have insurance coverage, reflecting the precarious healthcare financing situation of this workforce.

### 3.2 Hazard Perception, Safety Culture, and Injury Prevalence

**Table 3.2:** Hazard Perception, Safety Culture, and Injury Prevalence

| Variable | Category | Frequency (n) | Percentage (%) |
|---|---|---:|---:|
| **Hazard Awareness: Dust** | Yes | 288 | 76.6 |
| | No / Unsure | 88 | 23.4 |
| **Hazard Awareness: Noise** | Yes | 233 | 62 |
| | No / Unsure | 93 | 24.7 |
| **Machine Safety Guards** | Yes, machinery has adequate guards | 235 | 62.5 |
| | No, machinery lacks safety guards | 115 | 30.6 |
| | Not applicable to my job | 26 | 6.9 |
| **Ergonomic Autonomy (Breaks)** | Yes, can take breaks without problem | 141 | 37.5 |
| | Yes, but must inform supervisor | 175 | 46.5 |
| | No, breaks are only at fixed times | 50 | 13.3 |

| | | | |
|---|---|---|---|
| | No, there are very few breaks | 10 | 2.7 |
| **Confidence Reporting Safety** | Very confident | 116 | 30.9 |
| | Somewhat confident | 170 | 45.2 |
| | Not at all confident | 90 | 23.9 |
| **Time Off Due to Work Injury** | Yes | 216 | 57.4 |

To better understand the behavioral and organizational context of the Hayatabad Industrial Estate, workers were assessed on their perception of occupational hazards, workplace safety culture, and recent injury history (Table 3.2).

A considerable proportion of the workforce demonstrated a high awareness of specific physical hazards. The majority of workers (76.6%) correctly identified wood dust as a significant health risk. However, awareness regarding auditory hazards was notably lower, with only 62.0% recognizing the harmful effects of occupational noise. This leaves over a third of the workforce either unaware or unsure of noise-induced risks.

Despite general hazard awareness, the institutional safety culture revealed significant vulnerabilities. When asked about their confidence in reporting safety problems or hazards to management, only 30.9% of workers felt "Very confident." The largest segment (45.2%) felt only "Somewhat confident," and nearly one-quarter of the workforce (23.9%) reported feeling "Not at all confident" in bringing safety issues to light.

This precarious safety culture is reflected in a high rate of occupational trauma; more than half of the surveyed workforce (57.4%) reported having to take time off work due to a workplace injury.

Assessment of the physical and ergonomic workplace environment revealed critical gaps in primary safety infrastructure. Nearly one-third of the workforce (30.6%) reported operating machinery that lacked adequate safety guards or protective features. Regarding ergonomic autonomy and physical recovery, only 37.5% of workers felt they could take short breaks freely without issue. The largest proportion of the workforce (46.5%) was required to obtain explicit supervisory permission to rest, and 16.0% operated under rigid or highly restricted break schedules. This lack of ergonomic autonomy substantially compounds the cumulative musculoskeletal loading observed in this cohort

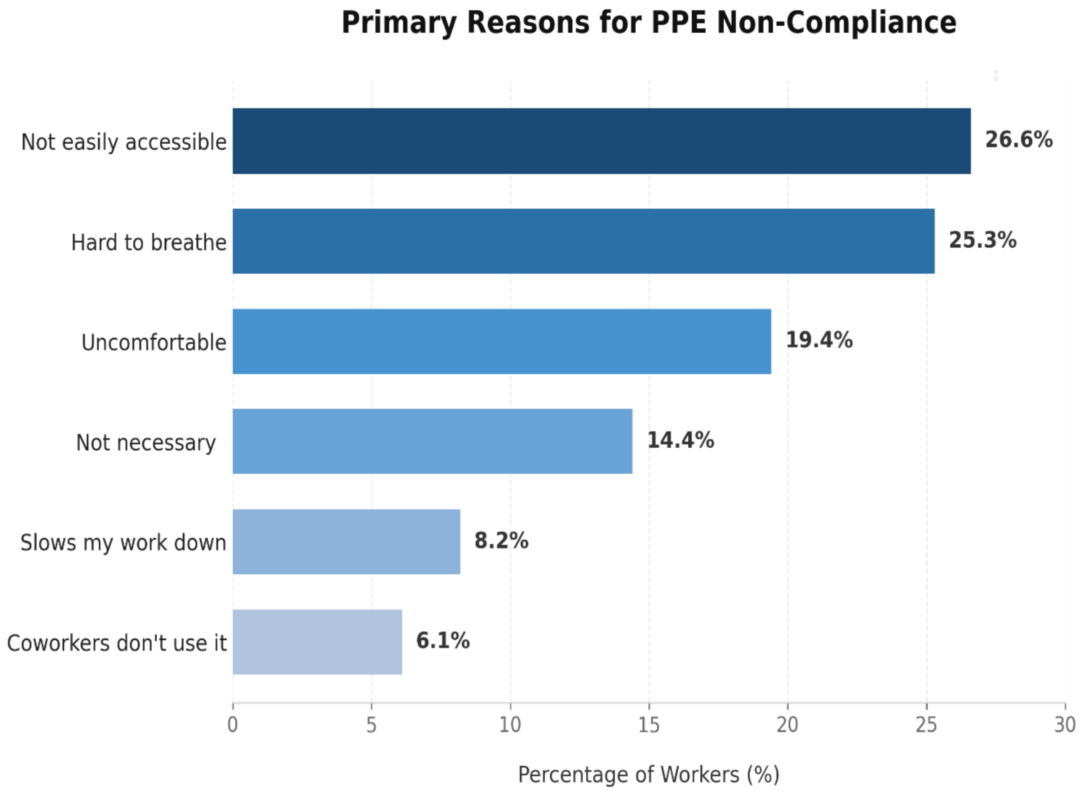


**Figure 4:** Primary Barriers to PPE Utilization Among Woodworkers

This bar chart illustrates the distribution of self-reported primary obstacles to the consistent use of Personal Protective Equipment (PPE) among the study cohort at the Hayatabad Industrial Estate.

### 3.3 Prevalence of Occupational Health Symptoms

Health symptoms were dichotomized into Low Symptom Burden (Never or Rarely reported) and High Symptom Burden (Sometimes or Often reported) to facilitate clinical interpretation and statistical analysis. The distribution of symptom frequencies across all six health domains measured in this study is presented in Figure 4.

Musculoskeletal pain emerged as the most prevalent occupational health problem, with 38.0% of workers (n = 143) reporting high-frequency symptoms. Respiratory cough was the second most prevalent condition (34.8%, n = 131), followed by headaches (27.1%, n = 102), eye irritation (23.9%, n = 90), skin problems (21.0%, n = 79), and tinnitus (17.6%, n = 66). The simultaneous co-occurrence of multiple symptoms within individual workers is consistent with the multi-hazard and cumulative exposure environment characteristic of wood processing settings, and underscores the importance of adopting comprehensive — rather than single-hazard — occupational health surveillance strategies.

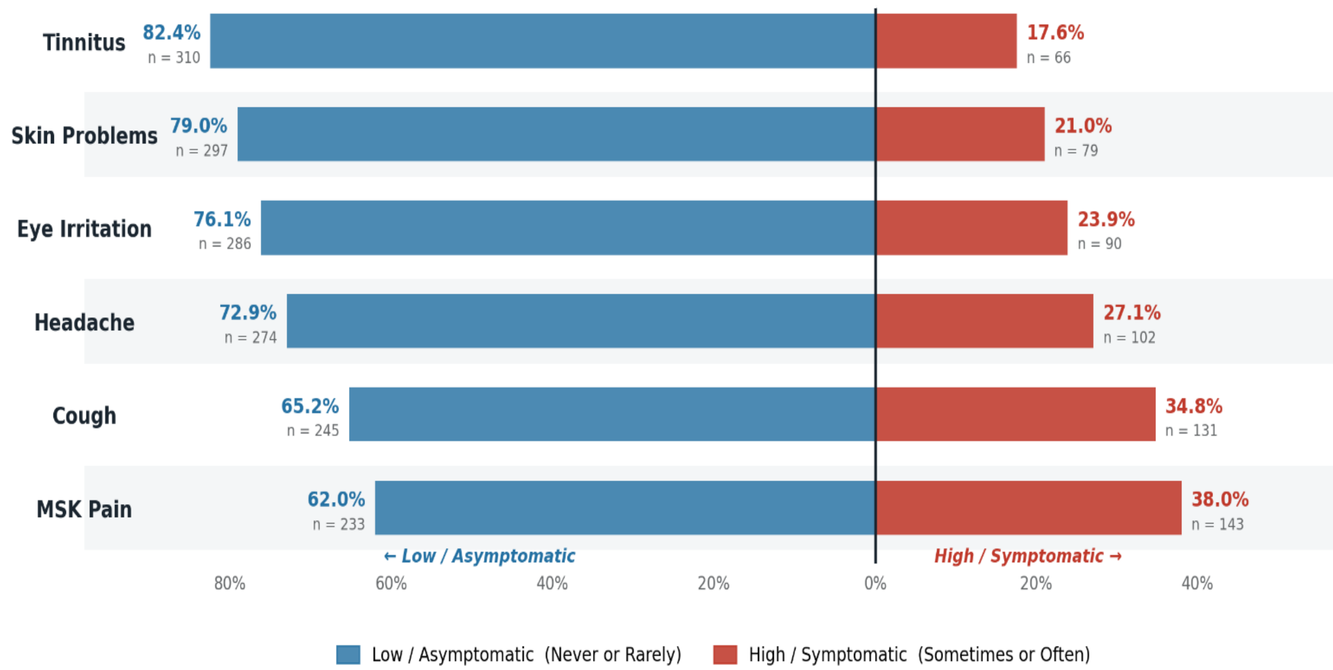


**Figure 5:** Prevalence of occupational health symptoms

## 3.4 Association Between PPE Usage and Health Symptoms

Chi-square analyses were performed to assess the statistical association between PPE usage frequency and each of the six health outcomes.

**Table 3.4: Chi-Square Analyses — Association Between PPE Usage Frequency and Health Outcomes**

| Health Outcome | PPE: Never | PPE: Sometimes | PPE: Always | Total | Symptom Frequency | $\chi^2$ | df | p |
|---|---|---|---|---|---|---|---|---|
| Respiratory Cough | 57 | 128 | 60 | 245 | Low | 34.733 | 2 | **< 0.001** |
| | 67 | 33 | 31 | 131 | High | | | |
| Eye Irritation | 84 | 126 | 76 | 286 | Low | 7.920 | 2 | **0.019** |
| | 40 | 35 | 15 | 90 | High | | | |
| Skin Problems | 85 | 134 | 78 | 297 | Low | 12.369 | 2 | **0.002** |
| | 39 | 27 | 13 | 79 | High | | | |
| Musculo-skeletal Pain | 66 | 102 | 65 | 233 | Low | 7.608 | 2 | **0.022** |
| | 58 | 59 | 26 | 143 | High | | | |
| Tinnitus | 88 | 139 | 83 | 310 | Low | 17.800 | 2 | **< 0.001** |
| | 36 | 22 | 8 | 66 | High | | | |
| Headache | 87 | 122 | 65 | 274 | Low | 1.244 | 2 | **0.537** |
| | 37 | 39 | 26 | 102 | High | | | |

PPE usage demonstrated statistically significant associations with five of the six health outcomes examined in this study, with the sixth — headache — falling short of statistical significance ($\chi^2 = 1.244$, $p = 0.537$). Among the significant associations, the strongest effect was observed for respiratory cough ($\chi^2 = 34.733$, $p < 0.001$), followed by tinnitus ($\chi^2 = 17.800$, $p < 0.001$), skin problems ($\chi^2 = 12.369$, $p = 0.002$), eye irritation ($\chi^2 = 7.920$, $p = 0.019$), and musculoskeletal pain ($\chi^2 = 7.608$, $p = 0.022$). Across all five significant outcomes, the distribution of symptom severity shifted notably toward lower frequency among workers reporting consistent PPE use, which lends face validity to the protective role of personal protective equipment in this occupational setting.

It is important, however, to interpret these bivariate associations with appropriate methodological caution. Chi-square analyses, by their nature, establish co-occurrence rather than causation, and the cross-sectional design of this study precludes any inference about temporality or directionality. A particular concern here is the possibility of reverse causality: workers assigned to higher-risk job roles — and thereby experiencing a greater symptom burden — may simultaneously be issued and observed wearing PPE more frequently as a direct consequence of institutional risk stratification or supervisory oversight. In other words, the relationship between PPE use and symptom severity may, in part, run in the opposite direction to what a straightforward protective interpretation would suggest. Additionally, unmeasured confounders such as duration of occupational exposure, workshop ventilation quality, and pre-existing health conditions could plausibly explain some portion of the observed variation. These limitations inherent to bivariate analysis underscore the necessity of the multivariate regression models presented in Section 3.6, which allow for simultaneous adjustment across relevant covariates and offer a more robust basis for causal inference.

## 3.5 Association Between Ventilation Status and Health Symptoms

**Table 3.5: Chi-Square Analysis — Association Between Workplace Ventilation and Respiratory/Eye Symptoms**

| Outcome | Ventilation | Never | Rarely | Sometimes | Often | $\chi^2$ (p-value) |
|---|---|---|---|---|---|---|

| | | | | | | |
|---|---|---|---|---|---|---|
| Cough | Present (n=315) | 128 | 90 | 69 | 28 | 22.92 (0.001)** |
| | Absent (n=59) | 7 | 18 | 25 | 9 | |
| Eye Irritation | Present (n=315) | 154 | 85 | 44 | 32 | 11.27 (0.080) ns |
| | Absent (n=59) | 23 | 23 | 12 | 1 | |

Workers in poorly ventilated workshops reported significantly higher frequencies of respiratory cough symptoms ($\chi^2 = 22.92$, $p = 0.001$), while the association between ventilation status and eye irritation did not achieve statistical significance ($\chi^2 = 11.27$, $p = 0.080$). The significant association with respiratory symptoms is consistent with the established pathophysiological role of inadequate ventilation in increasing airborne particulate and chemical concentrations within the breathing zone, thereby amplifying mucosal irritation and bronchoconstriction. The non-significant finding for eye irritation may reflect the multifactorial etiology of ocular symptoms in this setting, which may include direct mechanical particle contact not amenable to ventilation modification alone.

## 3.6 Association Between Job Role and Health Symptoms

Chi-square analyses were performed to assess the association between job role and each of the six health outcomes. Results are presented in Tables 3.6.1 through 3.6.3.

**Table 3.6.1: Chi-Square Analysis — Association Between Job Role and Health Outcomes**

| Health Outcome | $\chi^2$ | df | p-value | Significance |
|---|---|---|---|---|
| Respiratory Cough | 22.687 | 8 | 0.004 | Significant |
| Musculoskeletal Pain | 21.004 | 8 | 0.007 | Significant |
| Skin Symptoms | 15.180 | 8 | 0.056 | Borderline |
| Eye Symptoms | 4.638 | 8 | 0.795 | Not significant |
| Tinnitus | 5.067 | 8 | 0.750 | Not significant |
| Headache | 7.324 | 8 | 0.502 | Not significant |

Job role was significantly associated with respiratory cough ($\chi^2 = 22.687$, df = 8, $p = 0.004$) and musculoskeletal pain ($\chi^2 = 21.004$, df = 8, $p = 0.007$), reflecting the differential exposure profiles inherent to different woodworking occupations. Skin symptoms approached but did not reach conventional significance (p = 0.056), while eye symptoms, tinnitus, and headache showed no statistically significant variation by job role.

**Table 3.6.2: Job Role by Respiratory Cough Symptom Burden**

| Job Role | Low Symptom Burden | High Symptom Burden | Total |
|---|---|---|---|
| Machine Operator | 47 | 43 | 90 |
| Carpenter | 38 | 28 | 66 |
| Saw Operator | 52 | 21 | 73 |
| Helper | 43 | 14 | 57 |
| Supervisor | 14 | 12 | 26 |
| Spray Painter | 20 | 4 | 24 |
| Packing | 15 | 3 | 18 |
| Loader | 7 | 5 | 12 |
| Wood Polisher | 9 | 1 | 10 |
| **Total** | **245** | **131** | **376** |

Machine operators exhibited the highest absolute count of high cough burden (n = 43 of 90; 47.8%), followed by carpenters (n = 28 of 66; 42.4%) and supervisors (n = 12 of 26; 46.2%). Saw operators and spray painters showed comparatively lower proportional burdens despite working in dusty and chemically exposed environments, which may reflect confounding by age, PPE practices, or specific task patterns within those roles. The overall significant chi-square result (p = 0.004) confirms that respiratory symptom prevalence is not uniformly distributed across job categories but varies meaningfully with occupational function.

**Table 3.6.3: Job Role by Musculoskeletal Pain Symptom Burden**

| Job Role | Low Symptom Burden | High Symptom Burden | Total |
|---|---|---|---|
| Machine Operator | 62 | 28 | 90 |
| Carpenter | 38 | 28 | 66 |
| Saw Operator | 46 | 27 | 73 |

| | | | |
|---|---|---|---|
| Helper | 39 | 18 | 57 |
| Supervisor | 8 | 18 | 26 |
| Spray Painter | 13 | 11 | 24 |
| Packing | 13 | 5 | 18 |
| Loader | 5 | 7 | 12 |
| Wood Polisher | 9 | 1 | 10 |
| **Total** | **233** | **143** | **376** |

The distribution of musculoskeletal symptom burden across job roles reveals a striking finding: supervisors demonstrated the highest proportional prevalence of high MSK burden (18 of 26; 69.2%), substantially exceeding the study-wide average of 38.0%. This finding is counterintuitive if MSK disorders are assumed to be solely a function of acute physical exertion, and may instead reflect the cumulative effect of prolonged static postures, sustained workstation-based tasks, or extended working tenure characteristic of supervisory roles. Loaders also exhibited a notably high proportional MSK burden (7 of 12; 58.3%), consistent with the biomechanical demands of repetitive heavy manual lifting. Carpenters (42.4%) and spray painters (45.8%) showed above-average proportional burdens, while wood polishers had the lowest (1 of 10; 10.0%), possibly reflecting the relatively small sample size of that subgroup.

## 3.7 Screening of Independent Variables

To ensure the final multivariate models were both statistically robust and clinically meaningful, a strategic, two-step approach was adopted for data analysis. Rather than forcing all collected variables simultaneously into a regression model—a practice that risks model overfitting and masking true effects—we first conducted a comprehensive bivariate screening phase. During this stage, the standalone relationship between each independent predictor and the dichotomized health outcomes was systematically evaluated.

### 3.7.1 Bivariate Analysis of Categorical Predictors (Chi-Square Tests)

**Table 3.7.1: Summary of Significant Chi Square Results**

| Health Outcome | PPE Usage $\chi^2$(p) | Job Role $\chi^2$(p) | Ventilation $\chi^2$(p) | Smoking $\chi^2$(p) |
|---|---|---|---|---|
| Respiratory Cough | 34.733 (p <0.001*) | 22.687 (p=0.004*) | 16.836 (p<0.001*) | 6.769 (p=0.009*) |
| Eye Irritation | 7.920 (p=0.019*) | 4.638 (p=0.795) | 0.870 (p=0.647) | 0.616 (p=0.432) |
| Skin Problems | 12.369 (p=0.002*) | 15.180 (p=0.056) | 1.316 (p=0.518) | 0.392 (p=0.531) |
| MSK Pain | 7.608 (p=0.022*) | 21.004 (p=0.007*) | 0.341 (p=0.843) | 0.352 (p=0.553) |
| Tinnitus | 17.800 (p<0.001*) | 5.067 (p=0.750) | 1.237 (p=0.539) | 1.582 (p=0.209) |
| Headache | 1.244 (p=0.537) | 7.324 (p=0.502) | 0.648 (p=0.723) | 0.393 (p=0.531) |

To ensure the final multivariate logistic regression models were statistically robust, a bivariate screening phase was conducted. Table 3.7.1 summarizes the significant categorical associations identified in previous sections to establish which variables met the threshold for inclusion in the final regression analysis.

### 3.7.2 Bivariate Analysis of Continuous Predictors (Independent-Samples T-Tests)

Independent-samples t-tests were conducted to identify continuous variables demonstrating significant mean differences between workers with low and high symptom burden, prior to entry into binary logistic regression models. The complete t-test results for all six health outcomes and all continuous predictors are presented in the Supplementary Material (Table S1). Selected significant findings are summarized in Table 3.7 below.

**Table 3.7.2: Summary of Significant Independent-Samples T-Test Results**

| Outcome | Predictor | Mean (Low) | Mean (High) | t | p-value |
|---|---|---|---|---|---|
| *Cough* | Smoking status | 0.31 | 0.46 | 2.618 | 0.009 |
| | Age (years) | 33.58 | 36.77 | −2.592 | 0.010 |
| | Dust exposure score | 5.42 | 5.06 | 2.461 | 0.014 |

| | | | | | |
|---|---|---|---|---|---|
| | Standing hours/day | 5.96 | 5.11 | 3.414 | 0.001 |
| | PPE Usage | 2.01 | 1.73 | 3.582 | **0.001** |
| *Skin Problems* | Work duration (years) | 12.26 | 9.80 | 2.017 | 0.044 |
| | Dust exposure score | 5.42 | 4.82 | 3.553 | < 0.001 |
| | Noise exposure score | 6.09 | 5.46 | 2.265 | 0.024 |
| *MSK Pain* | Age (years) | 31.50 | 39.90 | −7.369 | < 0.001 |
| | Work duration (years) | 9.19 | 15.89 | −6.913 | < 0.001 |
| | Standing hours/day | 5.86 | 5.33 | 2.170 | 0.031 |
| | PPE usage score | 0.41 | 0.29 | 2.143 | 0.033 |
| *Tinnitus* | PPE usage score | 0.20 | 0.09 | 2.539 | 0.012 |
| *Headache* | Age (years) | 33.90 | 36.82 | −2.212 | 0.028 |
| | Noise exposure score | 6.14 | 5.47 | 2.606 | 0.010 |

The t-test results reveal several clinically and statistically compelling patterns. For respiratory cough, smokers demonstrated a markedly higher smoking proportion in the high-burden group (mean score 0.46 vs. 0.31; $p = 0.009$), while workers with high cough burden were significantly older (mean 36.77 vs. 33.58 years; $p = 0.010$). An unexpected finding was that higher dust exposure scores and more standing hours were paradoxically associated with lower cough burden — a pattern potentially attributable to survivorship bias, whereby workers in the most heavily dust-exposed or physically demanding roles who remain in the workforce may represent a selected, physiologically resilient sub-population.

For musculoskeletal pain, the magnitude of the mean difference in age (31.50 vs. 39.90 years; $t = -7.369$; $p < 0.001$) and work duration (9.19 vs. 15.89 years; $t = -6.913$; $p < 0.001$) between low- and high-burden groups represents the largest and most clinically significant contrasts observed across all t-test analyses in this study. These effect sizes — exceeding 8 years in age and 6.7 years in work tenure — provide robust support for the biological plausibility of cumulative musculoskeletal loading as the dominant driver of MSK symptom burden in this population.

For skin problems, workers with higher perceived dust exposure scores showed significantly greater skin symptom burden (mean 5.42 vs. 4.82; $p < 0.001$), consistent with established dermatological sensitization mechanisms from chronic wood dust contact. Tinnitus was significantly associated with lower PPE usage scores in symptomatic workers (0.09 vs. 0.20; $p = 0.012$), underscoring the direct protective function of hearing protection devices. For headache, both older age and higher noise exposure scores were significantly associated with greater symptom burden, supporting a cumulative neurological stress model.

## 3.8 Binary Logistic Regression Analyses

To resolve these mathematical limitations, the health outcomes were dichotomized into a binary scale (Low vs. High symptom burden). Binary logistic regression was therefore adopted as the final multivariate strategy. This approach satisfied all core statistical assumptions and allowed us to simultaneously control for potential confounders—such as age and cumulative work duration. The results of these models are presented as Adjusted Odds Ratios (AOR) with 95% Confidence Intervals (CI), providing stable, clinically interpretable measures of risk for each health outcome.

**Table 3.8.1: Binary Logistic Regression — Predictors of Respiratory Cough**

| Predictor | Crude OR (95%Cl) | p-value | AOR (95% CI) | p-Value |
|---|---|---|---|---|
| **Ventilation** | 2.45 (1.43 – 4.18) | 0.001* | 2.03 (1.14 – 3.61) | 0.016* |
| **Job Role** | 0.98 (0.89 – 1.08) | 0.713 | 1.00 (0.90 – 1.11) | 0.963 |
| **Smoking** | 1.89 (1.17 – 3.06) | 0.010* | 1.33 (0.78 – 2.26) | 0.301 |
| **PPE Usage** | 0.59 (0.44 – 0.79) | <0.001* | 0.63 (0.47 – 0.86) | 0.003* |
| **Age (years)** | 1.03 (1.01 – 1.04) | 0.011* | 1.01 (0.99 – 1.03) | 0.403 |
| **Dust Exposure Score** | 0.82 (0.70 – 0.96) | 0.015* | 0.91 (0.76 – 1.08) | 0.274 |

| | | | | |
|---|---|---|---|---|
| **Standing Hours** | 0.85 (0.77 – 0.94) | 0.001* | 0.91 (0.82 – 1.02) | 0.093 |

A comprehensive binary logistic regression analysis was conducted to identify the independent predictors of high respiratory cough burden among the study population. The multivariate model included ventilation status, job role, smoking, PPE usage, age, dust exposure, and standing hours. The overall model demonstrated a statistically significant improvement in fit over the null model ($chi^2 = 33.246$, df = 7, $p < 0.001$), explaining approximately 11.7% of the variance in cough symptom severity (Nagelkerke $R^2 = 0.117$). In the fully adjusted model, workplace ventilation emerged as a significant independent predictor of high cough burden: workers in poorly ventilated workshops had more than twice the odds of experiencing high respiratory cough burden compared to those in ventilated environments (AOR = 2.03; 95% CI: 1.14–3.61; p = 0.016). In contrast to preliminary models, the fully adjusted analysis revealed that occupational and behavioral factors superseded lifestyle factors as the primary drivers of respiratory health. Personal Protective Equipment (PPE) usage emerged as the strongest independent protective factor; for every one-unit increase in the frequency of PPE utilization, the odds of reporting a high cough burden decreased by 37% (AOR = 0.63; 95% CI: 0.47–0.86; p = 0.004). Additionally, prolonged standing hours showed a non-significant inverse trend with symptom severity (AOR = 0.91; 95% CI: 0.82–1.02; p = 0.093). Notably, while smoking, age, and total dust exposure exhibited significant bivariate associations with cough severity in crude analyses, none achieved statistical significance after multivariate adjustment, with ventilation and PPE usage accounting for the majority of explained variance (p > 0.05$). The adjusted odds ratio for smoking was reduced to a non-significant trend (AOR = 1.33; 95% CI: 0.78–2.26; p = 0.301), indicating that the raw effect of smoking is heavily confounded by the variance in protective workplace behaviors.

### *3.8.2 Predictors of Musculoskeletal Pain*

**Table 3.8.2: Binary Logistic Regression — Predictors of Musculoskeletal Pain**

| Predictor | Crude OR (95%Cl) | p-value | AOR (95% CI) | p-Value |
|---|---|---|---|---|
| **Age (years)** | 1.07 (1.05 – 1.09) | <0.001 | 1.05 (1.01 – 1.08) | 0.005 |
| **Total years in wood industry** | 1.08 (1.05 – 1.10) | <0.001 | 1.04 (1.00 – 1.07) | 0.057 |
| **PPE Usage Frequency** | 0.67 (0.51 – 0.89) | 0.006 | 0.74 (0.54 – 1.01) | 0.056 |
| **Standing Hours** | 0.91 (0.83 – 0.99) | 0.031 | 0.98 (0.89 – 1.09) | 0.732 |

A fully adjusted binary logistic regression was conducted to evaluate the predictors of high musculoskeletal (MSK) symptom burden. The multivariate model demonstrated exceptional robustness ($chi^2$ = 56.706, df = 4, p < 0.001), explaining 19.0% of the variance in MSK pain (Nagelkerke $R^2$ = 0.190), making it the most explanatory model fitted in this study. In contrast to respiratory outcomes, chronic biological factors dominated the MSK risk profile. Age emerged as the primary significant independent predictor: each additional year of life conferred a 5% increase in the odds of reporting severe MSK symptoms (AOR = 1.05; 95% CI: 1.01–1.08; p = 0.005). Cumulative occupational loading, measured by total years worked in the wood industry, demonstrated borderline significance (AOR = 1.04; 95% CI: 1.00–1.07; p = 0.057), reflecting the strong collinearity between natural aging and career tenure in this population. Notably, daily standing hours (p = 0.732) lost all predictive value after multivariate adjustment. Personal Protective Equipment (PPE) usage exhibited a marginally significant protective trend against MSK pain (AOR = 0.74; p = 0.056), though it did not cross the strict threshold for statistical significance."

### *3.8.3 Predictors of Skin Problems*

**Table 3.8.3: Binary Logistic Regression — Predictors of Skin Problems**

| Predictor | Crude OR (95%Cl) | p-value | AOR (95% CI) | p-Value |
|---|---|---|---|---|

| | | | | |
|---|---|---|---|---|
| **Dust Exposure Score** | 0.72 (0.59 – 0.87) | 0.001 | 0.72 (0.57 – 0.91) | 0.006 |
| **Total years in wood industry** | 0.97 (0.95 – 1.00) | 0.046 | 0.97 (0.94 – 1.00) | 0.022 |

A fully adjusted binary logistic regression was conducted to evaluate the independent predictors of high skin symptom burden. The multivariate model demonstrated a statistically significant improvement over the null model (Chi-square = 17.933, df = 3, $p < 0.001$), explaining 7.3% of the variance in dermatological complaints (Nagelkerke $R^2 = 0.073$). In the adjusted analysis, dust exposure and total years worked in the wood industry were identified as significant independent predictors. The dust exposure score demonstrated a persistent inverse relationship with symptom severity (AOR = 0.72; 95% CI: 0.57–0.91; $p = 0.006$), as did cumulative work duration, where each additional year in the industry was associated with a 3% reduction in the odds of high skin burden (AOR = 0.97; 95% CI: 0.94–1.00; $p = 0.022$). While ambient noise exposure was significant in the crude analysis (cOR = 0.88; $p = 0.025$), it lost all predictive value after controlling for dust and work duration in the multivariate model ($p = 0.793$).

### *3.8.4 Predictors of Headache*

**Table 3.8.4: Binary Logistic Regression — Predictors of Headache**

| Predictor | Crude OR (95%Cl) | p-value | AOR (95% CI) | p-Value |
|---|---|---|---|---|
| **Noise Exposure Score** | 0.87 (0.79 – 0.97) | 0.01 | 0.89 (0.80 – 0.99) | 0.026 |
| **Age (years)** | 1.02 (1.00 – 1.04) | 0.028 | 1.02 (1.00 – 1.04) | 0.077 |

A binary logistic regression was conducted to evaluate the independent predictors of high headache burden. The adjusted multivariate model demonstrated a statistically significant improvement over the null model (Chi-square = 9.803, df = 2, $p = 0.007$), explaining approximately 3.7% of the variance in headache frequency (Nagelkerke $R^2 = 0.037$). In the crude analysis, both age and noise exposure were significantly associated with headache outcomes. However, after multivariate adjustment, noise exposure emerged as the sole significant independent predictor (AOR = 0.89; 95% CI: 0.80–0.99; $p = 0.026$). The inverse adjusted odds ratio reflects the directionality of the specific ordinal scale used to capture noise exposure. While age demonstrated a positive association in the univariate model (cOR = 1.02; $p = 0.028$), it was attenuated to marginal significance once noise exposure was accounted for in the final model (AOR = 1.02; 95% CI: 1.00–1.04; $p = 0.077$).

#### ***3.8.5 Educational Attainment and PPE Compliance***

**Table 3.8.5: Spearman's Correlation — Educational Level and PPE Usage (N = 376)**

| **Variable Pair** | **rs** | **p-value** | **N** |
|---|---|---|---|
| Educational level — PPE usage frequency | 0.143 | 0.005 | 376 |

A statistically significant positive correlation was identified between workers' educational attainment and their self-reported PPE usage frequency ($r_s = 0.143$, $p = 0.005$), indicating that higher levels of formal education are associated with improved occupational safety compliance. While the effect size is modest, this finding has meaningful programmatic implications: it suggests that targeted educational and health literacy interventions — rather than PPE provision alone — may be essential components of effective compliance strategies in this workforce.

## 3.9 Quality of Life Impact

The occupational morbidity documented in this study translated directly into functional limitations outside the workplace. When asked if their work-related health problems affected their ability to perform household chores or enjoy their personal life, a striking majority reported a negative impact. Specifically, 61.2% of workers experienced a 'slight' reduction in their quality of life, while 6.9% reported that their occupational injuries 'significantly' impaired their personal and home life. Only 31.9% of the workforce reported that their home life remained completely unaffected by their occupational health status.